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Acids and Bases

Acid: The compound which contain hydrogen atom and donate hydrogen ion (H^+) ion when dissolved In water is called acid. e.g. HCl, H_2SO_4, HNO_3,H_2CO_3 etc.

But compound which contain hydrogen atom such as methane (CH_4), ethane(C_2H_6) and propane (C_3H_8) are not acid because it cannot produces hydrogen (H^+) ion when dissolved in water.

Classification:

 # According to strength:

i) Strong acid: The compound which donates hydrogen ion completely when dissolved in water is called strong acid. e.g. *HCl*, H_2SO_4, HNO_3etc.

$$HCl \longrightarrow H^+ + Cl^-$$
$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

ii) Weak acid: The compound which donates hydrogen ion partially when dissolved in water is called weak acid. e.g. Carbonic acid (H_2CO_3), acetic acid ($CH_3 - COOH$) and most of the organic acid.

$$H_2CO_3 \rightleftharpoons 2H^+ + CO_3^{2-}$$
$$CH_3 - COOH \rightleftharpoons CH_3 - COO^- + H^+$$

 # According to source:

i) Inorganic/ mineral acid: HCl, H_2SO_4, HNO_3,H_2CO_3 etc.

ii) Organic acid: Formic acid ($H - COOH$), acetic acid ($CH_3 - COOH$), oxalic acid ($HOOC - COOH$), etc.

 # Depending on structure:

I) Hydracids: The acid which contain hydrogen and any other nonmetal atom but not oxygen is called hydracids. Example: HI, HCl, HBr, HF etc.

ii) Oxyacids: The acid which contain oxygen and one or more nonmetal atom is called oxyacids. Example: HNO_3, H_2SO_4 etc.

 # Depending on acid ionization or dissociation constant:

a) Very weak acid:
- HClO (Hypo chlorous acid)
- HBrO (Hypo bromous acid)
- H_4SiO_4 (Silicic acid)
- H_3BO_3 (Boric acid)

b) Weak acid:

- H_2CO_3 (Carbonic acid)
- HNO_2 (Nitrous acid)
- H_3PO_4 (Phosphoric acid)
- H_3PO_3 (Phosphorous acid)
- H_2SO_3 (Sulphurous acid)
- $HClO_2$ (Chlorous acid)

c) Very strong acid**:**

- $HClO_4$ (Perchloric acid)
- $HMnO_4$ (Permanganic acid)

d) Strong acid:

- HCl (Hydrochloric acid)
- HNO_3 (Nitric acid)
- H_2SO_4 (Sulfuric acid)
- $HClO_3$ (Chloric acid)
- H_2SeO_4 (Selanic acid)

Base: The compound which contain oxide (O^{2-}) or hydroxide (OH^-) ion and react with acid to form salt and water is called base. e.g. Sodium oxide(Na_2O), Sodium hydroxide (NaOH), Calcium oxide (CaO), Calcium hydroxide $Ca(OH)_2$ etc.

Alkali: The compound which contain hydroxyl group ($-OH$)and donate hydroxide ion (OH^-) when dissolved in water is called alkali. e.g. NaOH, KOH, LiOH, $Ca(OH)_2$etc.

"**All alkali is base but all bases are not alkali."**

But in spite of hydroxyl group ($-OH$) present in these compound such as methanol (CH_3OH), ethanol (C_2H_5OH) are not base because it does not yields hydroxyl ion (OH^-) when dissolved in water.

Classification:

 # Depending on strength:

i) Strong base: The compound which donates hydroxyl ion completely or fully is called strong base. e.g.

- NaOH (sodium hydroxide)
- KOH (potassium hydroxide)
- $Ba(OH)_2$ (barium hydroxide)
- LiOH - lithium hydroxide
- RbOH - rubidium hydroxide
- CsOH - cesium hydroxide

ii) Weak base: The compound which donates hydroxyl ion partially is called weak base. e.g.

NH_3 (ammonia)
CH_3NH_2 (methylamine)
C_5H_5N (pyridine)
NH_4OH (ammonium hydroxide)

Different concept of acids & bases

Arrhenious concept of acids and bases:

An acid is a hydrogen containing substance which yields hydrogen ion(H^+)(i.e. hydronium ion,H_3O^+) when dissolved in water.

$$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$$

A base is a substance which contain hydroxyl group($-OH$)and produces hydroxyl ion (OH^-)when dissolved in water.

$$NaOH \longrightarrow Na^+ + OH^-$$

Bronsted-Lowery concept of acids and bases:

An acid is a substance that can give up proton to another species.
A base is a substance that can accept protons.

Example: When ammonia is dissolved in water, the reaction takes place in the following way:

$$H_2O + :NH_3 \rightleftharpoons NH_4^+ + OH^-$$

Here, water gives up the proton, so it is acid and ammonia accepts the proton so it is base. In the reverse reaction, NH_4^+ donates a proton to OH^- so NH_4^+ion is the acid and $OH^-$is the base.

+ H^+

- H^+

$$H_2O + :NH_3 \rightleftharpoons NH_4^+ + OH^-$$

+ H^+

- H^+

A conjugate acid –base pair consists of two species in acid –base reaction, one acid and one base that differ by the loss or gain of a proton. The acid in such a pair is called the conjugate acid of the base, where as the base is the conjugate base of the acid.

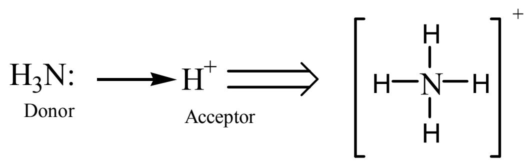
In the following equations label each species as an acid or a base. Show the conjugate acid-base pair.

i) HSO_4^- (aq) + HI(aq) $\rightleftharpoons$ H_2SO_4(aq) + $I^-(aq)$
ii) HCO_3^-(aq) + CN^-(aq) $\rightleftharpoons$ CO_3^{2-}(aq) + HCN(aq)

 # Lewis concept of acids and bases:

A Lewis acid is a species that can form a covalent bond by accepting an electron pair from another species.

A Lewis base is a species that can form a covalent bond by donating an electron pair to another species. e.g.



In the following reactions, identify the Lewis acid and Lewis base.

a) $Ag^{+} + 2NH_3 \rightleftharpoons Ag(NH_3)_2^{+}$

b) $B(OH)_3 + H_2O \rightleftharpoons B(OH)_4^{-} + H^{+}$

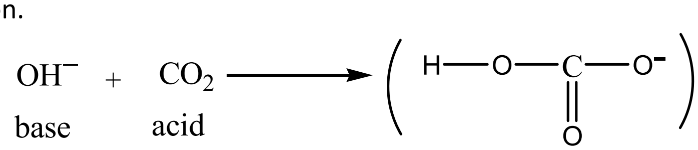
Q-1: How does a strong acid differ from a weak acid?

Classify each of the following as acidic, basic or neutral properties:

01	Oven cleaner	06	Soap solution	11	Unripe mangoes
02	Tomatoes	07	Carbonated beverages	12	Sodium hydroxide solution
03	Household bleach	08	Curd	13	Baking powder in water
04	Lemon Juice	09	Gastric fluid	14	Vinegar in water
05	Ant's Sting	10	Lime water	15	Glucose in water

Q-2: Self-contained environments, such as that of a space station require that the carbon-di oxide exhaled by people be continuously removed. This can be done by passing the air over solid alkali hydroxide ion. What ion is produced by the addition of OH^{-}ion to CO_2? Use the Lewis concept to explain this.

Answer: The hydroxide ion acts as a base and donates a pair of electrons on the O atom, forming a bond with $CO_2$to give the HCO_3^{-}ion.



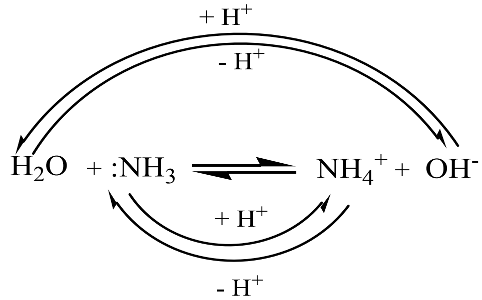
Q-3: Aqueous solution of ammonia,NH_3, were once thought to be solutions of an ionic compound (ammonium hydroxide, NH_4OH) in order to explain how the solutions could contain hydroxide ion. Using the Bronsted-Lowery concept, show how NH_3 yields hydroxide ion in aqueous solution without involving the species NH_4OH. Indicate the species that are conjugates of one another.

Answer:

When ammonia is dissolved in water, the reaction takes place in the following way:

$$H_2O + :NH_3 \rightleftharpoons NH_4^{+} + OH^{-}$$

Here, water gives up the proton, so it is acid and ammonia accepts the proton so it is base. In the reverse reaction, NH_4^{+} donates a proton to OH^{-} so NH_4^{+}ion is the acid and $OH^{-}$is the base.



Q-4: Natural gas frequently contains hydrogen sulfide,H_2S is removed from natural gas by passing it through aqueous ethanolamine, HO$CH_2CH_2NH_2$(an ammonia derivative),which reacts with the hydrogen sulfide. Write an equation for the reaction.

i) Identify each reactant as either a Lewis acid or a Lewis base. Explain how you arrived at your answer.

ii) Identify each reactant and product as an acid or a base. Indicate the species that are conjugates of one another

Answer:

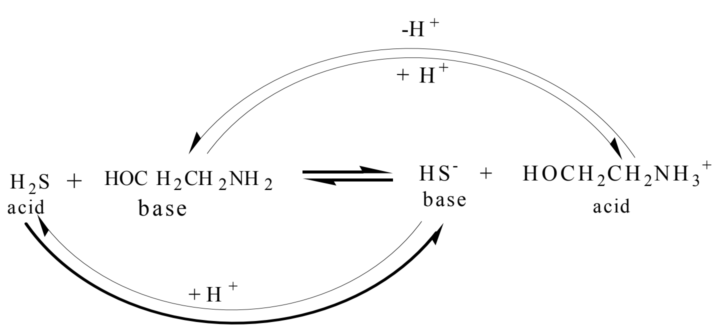
i) $\underset{\text{Lewis acid}}{H_2S} + \underset{\text{Lewis base}}{HOCH_2CH_2NH_2} \longrightarrow HS^{-} + HOCH_2CH_2NH_3^{+}$

The H^{+} ion from H_2S accepts a pair of electrons from the N atom in $HOCH_2CH_2NH_2$.

$H_2S \longrightarrow H^{+} + HS^{-}$

$\underset{\text{Electron pair acceptor}}{H^{+}} + \underset{\text{electron pair donor}}{HOCH_2CH_2\ddot{N}H_2} \longrightarrow HOCH_2\,CH_2\,NH_3^{+}$

ii) On the left, H_2S is proton donor, on the right, $HOCH_2CH_2NH_3^{+}$is the proton donor. The proton acceptors are $HOCH_2CH_2NH_2$ and HS^{-}.Once the proton donors and acceptors are identified, the acids and bases can be labeled.



In this reaction, H_2S and $HS^{-}$are a conjugate acid-base pair, as are $HOCH_2CH_2NH_2$ and $HOCH_2CH_2NH_3^{+}$

Q-5: "In spite of hydrogen atom present in ammonia (NH_3), does not exist as a acid but it act as a base"- explain

Answer: When ammonia dissolved in water it produces ammonium hydroxide (NH_4OH) base which changes red litmus to blue. So ammonia is base. Again base react with acid to form salt and water but ammonia react with acid to form only salt.

$$NH_3 + HCl = NH_4Cl$$

For this reason ammonia is called base.

According to Bronsted-Lowery concept, a base is a substance that can accept proton but ammonia accept proton(H^+) easily because it has one lone pair electron. So it is base.

Q-6: Phosphorous acid, H_3PO_3 and phosphoric acid,H_3PO_4 have approximately the same acid strengths. From this information and noting the possibility that one or more hydrogen atoms may be directly bonded to the phosphorous atom, draw the structural formula of phosphorous acid. How many grams of sodium hydroxide would be required to completely neutralize 1.00 g of this acid?

Q-7: When aqueous solution of CuSO_4 is added to the aqueous solution of $(NH_4)_2SO_4$, the acidity of $(NH_4)_2SO_4$ solution increases. Explain.

Answer: When CuSO_4 Solution is added to$(NH_4)_2SO_4$, the following reaction takes place:

$$CuSO_4 + (NH_4)_2SO_4 \longrightarrow [Cu(NH_3)_4]SO_4 + 2H_2SO_4$$

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

Due to production of H^+ions, the acidity of aqueous solution of $(NH_4)_2SO_4$ increases.

Acid strength:

The strength of halogen acid depends on the size of atom X. The larger atom X, the weaker is the bond and greater the acid strength. In going down a column of elements of the periodic table, the size of the atom X increases, the H — X bond strength decreases and the strength of the binary acid increases. So the strength of halogen acids is as follows:

$$HF < HCl < HBr < HI$$

Q-8: Which of the following acids is the strongest acid: ClO_3(OH), ClO_2(OH), SO(OH)$_2$, SO_2(OH)$_2$?

Answer:

Given any of the above acids can be written as $HClO_4$, $HClO_3$, H_2SO_3, H_2SO_4. The oxidation number of central atom of these acids is $+7$, $+5$, $+4$ and $+6$ respectively. Since the oxidation number of $HClO_4$ is the highest, this acid is the strongest acid.

The $\boldsymbol{P^H}$ of a solution:

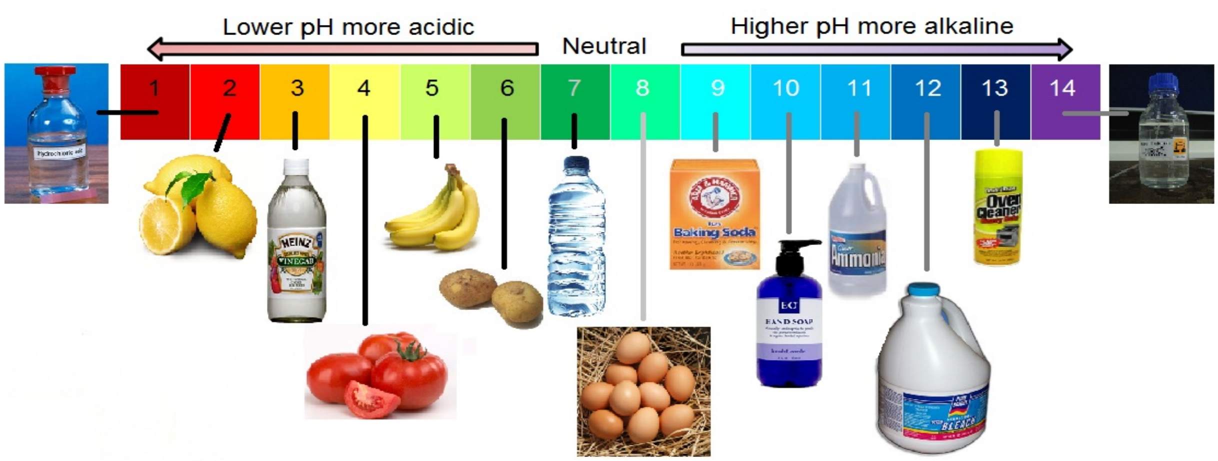
P^H which is defined as the negative logarithm of molar hydrogen ion concentration.

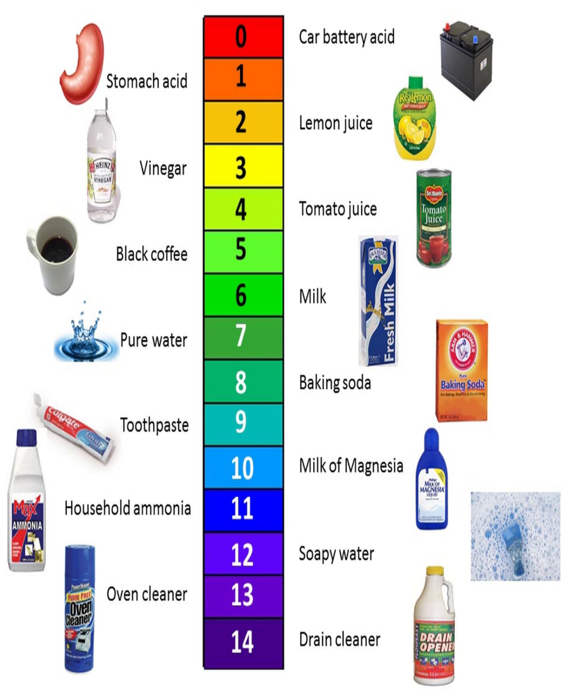
$$P^H = -\log_{10}[H^+]$$

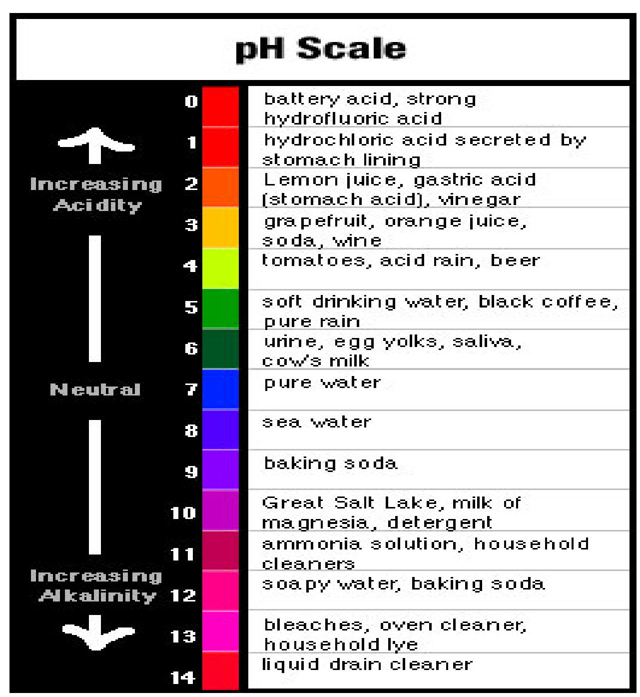
The pH system is an international way to indicate the acidity or alkalinity (= strength of a base). This scale ranges from 0 to 14.The lower the pH value, the more acidic the solution. The higher the pH, the more alkaline the product. Some examples of pH values:

Sulfuric acid: pH 1
Citric acid: pH 2
Acetic acid: pH 3
Water: pH 7
Ammonia: pH 11

The pH scale can be shown graphically as in the picture below:







Problem:

i) A sample of orange juice has hydrogen ion concentration of 2.9×10^{-4}M. What is the P^H ? Is the solution acidic?

ii) What is the $P^H$of a sample of grastic juice whose hydrogen ion concentration is 0.045M?

iii) The $P^H$of human arterial blood is 7.40. What is the hydrogen ion concentration?

iV) A brand of carbonated beverage has a $P^H$of 5.16. What is the hydrogen ion concentration of the beverage?

V) Morphine is a narcotic that is used to relieve pain. A solution of morphine has a pH of 9.61 at 25^0C.What is the hydrogen ion concentration?

Vi) The $P^H$of a cup of coffee at 25^0C was found to be 5.12. What is the hydrogen ion concentration?

Vii) A sample of vinegar has hydroxide ion concentration of 2.5×10^{-3}M. What is the $P^H$of the vinegar?

Viii) Which of the following P^Hvalues indicate an acidic solution at25°C? Which are acidic and which are neutral?

a) 4.6 b) 2.5 c) 1.6 d) 10.5

Buffer:

A buffer is a solution characterized by the ability to resist changes in P^H when limited amount of acid or base are added to it.

Buffers contain either a weak acid and its conjugate base or a weak base and its conjugate acid. Thus a buffer solution contains both an acid species and a base species in equilibrium. e.g.

H_2CO_3 + HCO_3Na, CH_3COOH + CH_3COONa, NH_4OH + NH_4Cl

15.100 A 2.500-g sample of a mixture of sodium carbonate and sodium chloride is dissolved in 25.00 mL of 0.798 *M* HCl. Some acid remains after the treatment of the sample.

a. Write the net ionic equation for the complete reaction of sodium carbonate with hydrochloric acid.

b. If 28.7 mL of 0.108 *M* NaOH were required to titrate the excess hydrochloric acid, how many moles of sodium carbonate were present in the original sample?

c. What is the percent composition of the original sample?

Question (1): Define the term "pH"; what does" pH" stand for?
Answer: P^{H}which is defined as the negative logarithm of molar hydrogen ion concentration.

$$P^{H} = -\log[H^{+}]$$

pH stands for: Power of hydrogen ion concentration, 'p' for power and 'H' for H^{+} ion concentration.

Question (2): What is 'pH' scale? Explain briefly.
Answer: The strength of an acid or a base is expressed in terms of hydronium ion concentration. This is expressed on a scale known as 'pH' scale. It is a 14 point scale; i.e., it has values ranging from 0 to 14, indicating the value of negative logs of H^{+} ion concentration of the solution. Some important benchmark value$_s$ in the pH scale are: pH = 7 indicate neutral solutions e.g., aqueous solutions. pH > 7 to 14 indicates alkaline solutions and pH < 7 to 0 indicate acidic solutions

Question (3): What is the 'pH' of pure water and that of rain water? Explain the difference.
Answer: The pH of pure water is seven. Rain water is slightly acidic because as rain drop fall, the carbon dioxide in the air dissolves with drops to form very weak carbonic acid. Accordingly, rain water has a pH that is slightly below 7

Question (4): What is the pH of solution 'A' which liberates CO_2 gas when reacts with a carbonate salt? Give the reason?
Answer: The pH of solution 'A' is less than 7 because it reacts with carbonate salt to form carbon di-oxide gas. In this case it is acid.
The reaction between carbonates salts and acid (A) (suppose hydrochloric acid) is given below:

$$CaCO_3 + HCl = CaCl_2 + H_2O + CO_2$$

Question (5): What is the pH of solution 'B' which liberates NH_3 gas when reacts with an ammonium salt? Give reason?
Answer: The pH of solution 'B' is less than 7 because 'B' is an alkali as it liberates NH_3 gas.The reaction occurs between ammonia salt and alkali is given below:

$$NH_4Cl + NaOH = NaCl + NH_3 + H_2O$$

Question (16): How do you increase or decrease the pH of pure water?
Answer: By adding a few drops of alkali to pure water, it's pH increases; and by adding a few drops of an acid decreases the pH of pure water.

Question (6). A milkman adds a very small amount of baking soda to fresh milk.
(a) Why does he shift the pH of the fresh milk from 6 to slightly alkaline?
(b) Why does this milk take a long time to set as curd?
(c) Fresh milk has a pH of 6. How does the pH will change as it turns to curd? Explain your answer

Answer: (a) By adding small amount of baking soda, the milkman shifts the pH of the fresh milk from 6 to slightly alkaline so that he can keep it for longer time as the milk in alkaline condition, does not set curd easily.

(b) Since, this milk is slightly basic than the usual milk, acids produced due to bacterial actions in it are neutralized by the base. Therefore, it takes longer time to set as curd.

(c) pH will decrease as it turns to curd because curd is acidic due to the presence of lactic acid.

Question (7): (a) Write the name given to bases that are highly soluble in water. Give an example.
(b)How is tooth decay related to pH? How can it be prevented?
(c)Why does bee sting cause pain and irritation? Rubbing of baking soda on the sting

area gives relief. How?

Answer.(a) Alkali, e.g. NaOH (Sodium hydroxide).
(b) Lower the pH, more will be tooth decay. Acid reacts with $Ca_3(PO_4)_2$ and cause tooth decay.
It can be prevented by brushing teeth after every meal.
(ic) It is due to formic acid. Sodium hydrogencarbonate (Baking soda) neutralises formic acid giving relief.